

Lab 9

Introduction to Functions in C++

Objectives

The objective of this lab is to:

- Introduction to Function
- Prototype of Function
- Function declaration
- Function Call
- Syntax
- Examples

Introduction

Functions let you chop up a long program into named sections so that the sections can be reused throughout the program. Function has a name identifier, accepts parameters and returns a result. C++ functions can accept an unlimited number of parameters. In general, C++ does not care in what order you put your functions in the program, so long as the function name is known to the compiler before it is called. We have been consistently using library functions in the programs such as `getch()` and `clrscr()`. We just simply use these functions without knowing what's inside these functions. We just call them from our program with some parameters inside the brackets and it does the work for us. We don't know its implementation. When we actually call these functions, the code inside these functions executes and at the end it transfers the control back to the program from where the function was called. In this lab we would learn how to make our own functions which can perform certain tasks. Function is differentiated from a simple variable, by the parenthesis just after the name as `main()`. It just indicates to the compiler that it's a function. The most common function that you have encountered is `main()` which you have been implementing throughout the lab. `main()` is the function whose name identifier is `main` with no parameters passed to it. Anything inside brackets are the parameters passed to the function. In case of multiple parameters, each of them are separated with a comma(,) in between. Parameters are written as the comma separated list within brackets just after the name of function. In `main()` function there are no parameters passed to it as the brackets after the name is empty. As you know when we use `int main()` in our program then we return an integer from the main function. Similarly when we define our own functions in the program, that function should return appropriate value according to the data type which is given before the name of function when we give its definition or implementation, just as we return an integer when we use `int main()` and nothing in case of `void main()`. C++ assumes that every function will return a value. If the programmer wants a return value, this is achieved using the return statement. If no return value is required, none should be used when calling the function.

Porotype of Function

The prototype of a function is a way of describing the parameters (also called as arguments of function) and parameter types with which a legal call to the function can be made. It contains the name of the function, its parameters and their type, and the return value.

Syntax:

```
returnType functionName(List of parameters separated by commas);
```

Example:

```
int main();  
  
float divide(int a, int b);  
  
void square(float x);
```

Here names of the functions are main, divide and square which take 0, 2 and 1 parameters respectively.

Example:

In this example main function just calls the function named asterisks which just prints the line of ten asterisks.

```
#include using namespace std;  
  
void asterisk(); // function prototype  
  
int main() {  
    asterisk(); // function call  
    return 0;  
}  
  
// function definition  
void asterisk()  
{ for (int i = 0; i < 10; i++) { cout << "*"; } }
```

The asteriks() function in the previous example does not take any argument and does not return anything represented by empty braces () after function name and void before function name respectively.

Example:

Let's go one step ahead, function asterisks (int a) with a single argument.

```
#include using namespace std;

void asterisk(int n);

// function prototype

int main() {
    asterisk(12); // function call
    return 0; }

// function definition
void asterisk(int num)
{
    for (int i = 0; i < num; i++)
    { 15 cout << "*"; 16 } 17
}
```

This function would take an integer number as an argument and prints the number of asterisks equal to the number given as argument to the function in straight line. As it is given in function prototype, compiler expects function definition should take single argument which should be of integer data type. Due to this in function calling you need to give an integer as argument (parameter) of a function.

Example:

Now an example of a function with two parameters.

```
#include
using namespace std;
int add(int, int);
// function prototype
int main() {
    int result;
    // function call
    result = add(3, 5);
    cout << result;
    return 0; }
// function definition
int add(int num1, int num2) {
    return num1 + num2;}
```

Lab Tasks:

Task 1: Function Introduction and Declaration (greet)

- Declare a function called "greet" that takes no parameters and returns nothing.
- Inside the function, display a greeting message of your choice.
- Call the function from the main program to execute the greeting.

```
#include <iostream>           // Include input-output library
using namespace std;         // Use standard namespace

void greet() {                // Declare function with no parameters
                                and no return
    cout << "Welcome to C++ Programming Lab!" << endl;
                                // Display greeting message
}

int main() {                  // Main function starts
    greet();                  // Call the greet function
    return 0;
}    cout << endl;
    }
    return 0;                // End of program
}
```

Task 2: Basic Function Declaration and Call (sayHello)

- Declare a function called "sayHello" that takes no parameters and returns
- nothing.
- Inside the function, display the message "Hello, World!".
- Call the function from the main program to execute the greeting.

```
#include <iostream>           // Include input-output library
using namespace std;         // Use standard namespace

void sayHello() {             // Declare function
    cout << "Hello, World!" << endl; // Print message
}

int main() {                  // Main function
    sayHello();               // Call function
    return 0;                 // End program
}
```

Task 3: Array Input, Function for Sum & Average

- Write a C++ program that:
- Takes 10 integer values as input from the user and stores them in a 1D array.
- Creates a function named calculateSum that:
- Accepts the array and its size as parameters
- Returns the sum of all elements
- In the main() function:
- Call the function
- Calculate and display the average of the array elements
- Constraints:
- Use only one-dimensional array
- Use function for sum calculation

```

#include <iostream>           // Include input-output library

using namespace std;         // Use standard namespace

int calculateSum(int arr[], int size) { // Function to calculate sum

int sum = 0;                  // Initialize sum variable

for(int i = 0; i < size; i++) { // Loop through array

    sum += arr[i]; }          // Add each element to sum

return sum; }                // Return total sum

int main() {                  // Main function

int arr[10];                  // Declare array of size 10

int sum;                      // Variable to store sum

float average;                // Variable to store average

cout << "Enter 10 numbers:\n"; // Prompt user

for(int i = 0; i < 10; i++) { // Loop for input

    cin >> arr[i]; }          // Store input in array

sum = calculateSum(arr, 10);   // Call function and store result

average = (float)sum / 10;     // Calculate average

cout << "Sum = " << sum << endl; // Display sum

cout << "Average = " << average << endl; // Display average

return 0; }                   // End program

```